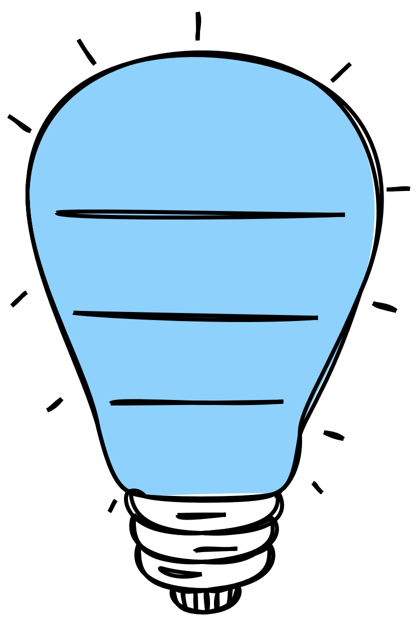
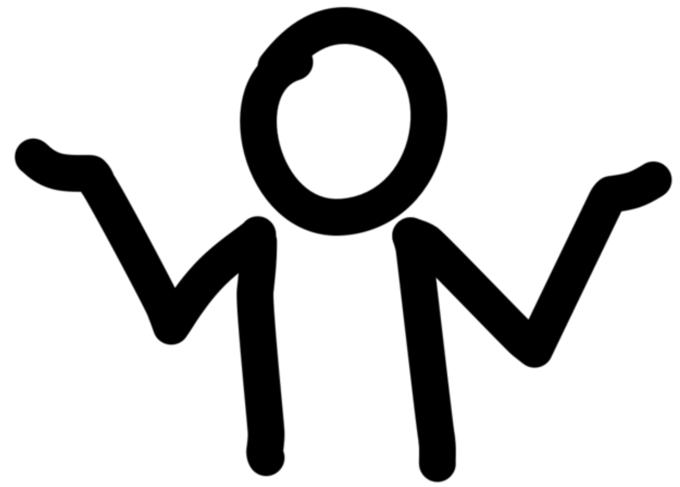
How to choose a PSU [Power Supply]

600W?



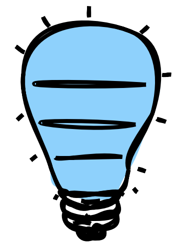
800W?

1000W?



80 Plus?

How many watts?



Component	Power Requirement	How to Find?
CPU (Intel Core i7-13700K)	125 W	Product Specs
GPU (NVidia RTX 4060)	115 W	Product Specs
RAM (64 GB DDR5)	25 W	Estimate
Motherboard (ATX)	70 W	Estimate
SSD Storage Drive (1TB)	15 W	Estimate
HDD Storage Drive (1 TB, 10k RPM 3.5”)	15 W	Estimate
Estimated Power Requirement	365 W	

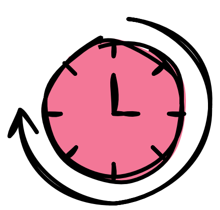
Use an online calculator or go by the GPU requirements

NVIDIA GPU's

GPU	Min Recommended PSU (W)	Required Connectors
RTX 4090	850	3x PCIe 8-pin cables (adapter in box) or 1x 450 W or greater PCIe Gen 5 cable.
RTX 4080	750	3x PCIe 8-pin cables (adapter in box) or 1x 450 W or greater PCIe Gen 5 cable.
RTX 4070 Ti	700	2x PCIe 8-pin cables (adapter in box) or 300 W or greater PCIe Gen 5 cable.
RTX 4070	650	2x PCIe 8-pin cables (adapter in box) or 300 W or greater PCIe Gen 5 cable. Certain models may use 1x PCIe 8-pin cable.
RTX 4060 Ti	550	1x PCIe 8-pin cables (adapter in box) or 300 W or greater PCIe Gen 5 cable. Certain models may use 1x PCIe 8-pin cable.
RTX 4060	550	1x PCIe 8-pin cables (adapter in box) or 300 W or greater PCIe Gen 5 cable. Certain models may use 1x PCIe 6-pin cable or 1x PCIe 8-pin cable.
RTX 3090 Ti	850	3x PCIe 8-pin cables (adapter in box) OR 450W or greater PCIe Gen 5 cable
RTX 3090	750	2x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3080 Ti	750	2x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3080	750	2x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3070 Ti	750	2x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3070	650	1x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3060 Ti	600	1x PCIe 8-pin (adapter to 1x 12-pin included)
RTX 3060	550	1x PCIe 8-pin
RTX 3050	550	1x PCIe 8-pin

AMD GPU's

GPU	Min Recommended PSU (W)	Required Connectors
RX 7900 XTX	800	2x PCIe 8-pin cables
RX 7900 XT	750	2x PCIe 8-pin cables
RX 7800 XT	700	2x PCIe 8-pin cables
RX 7700 XT	700	2x PCIe 8-pin cables
RX 7600	550	1x PCIe 8-pin cables
RX 6950 XT	850	2x PCIe 8-pin cables
RX 6900 XT	850	2x PCIe 8-pin cables
RX 6800 XT	750	2x PCIe 8-pin cables
RX 6800	650	2x PCIe 8-pin cables
RX 6700	600	1x PCIe 8-pin cables



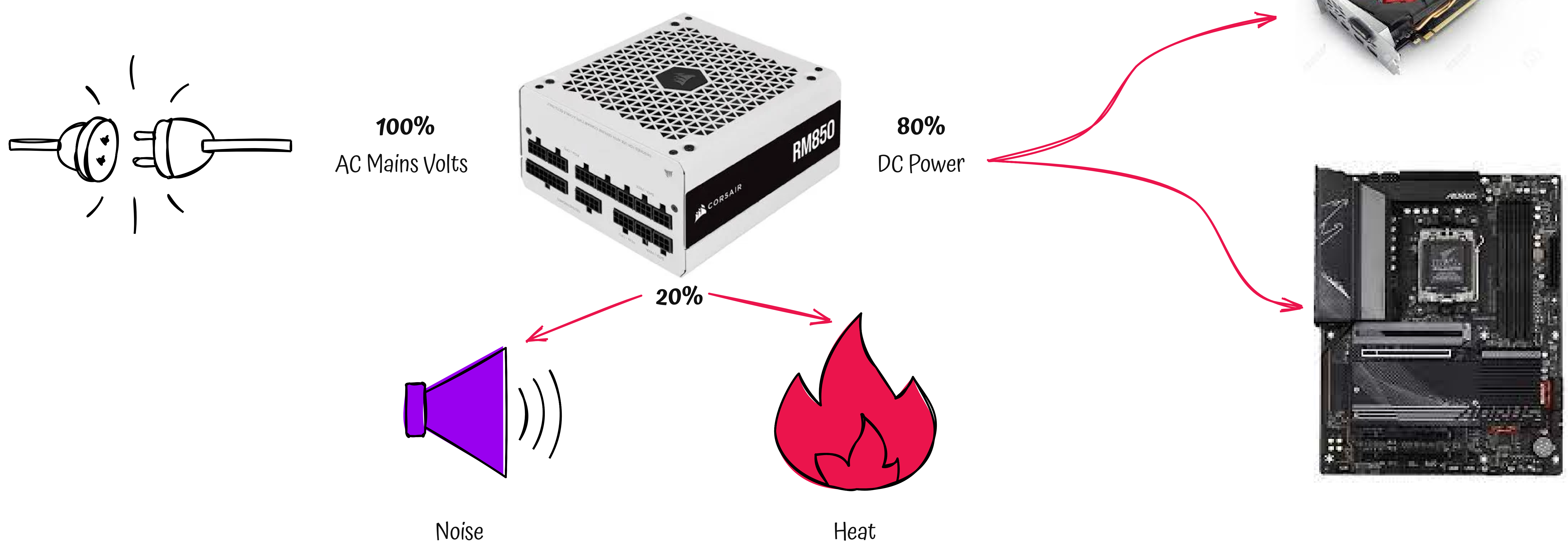
Overclocking uses more power

Any PSU with this badge is 80% efficient
What is 80% efficient?

Certification	115V	230V (EU)
80 PLUS	80%	82%
80 PLUS Bronze	82%	85%
80 PLUS Silver	85%	87%
80 PLUS Gold	87%	89%
80 PLUS Platinum	89%	90%
80 PLUS Titanium	90%	91%

Efficiency Certification @ 100% Load

Higher certification does not certify longer lifetime
Better components and manufacture may give that



ATX

The different PSU Sizes

Most cases, except the specialist smaller form factors will take an ATX PSU

Specialist Sizes

PSU Size	Width (mm)	Height (mm)	Depth (mm)
ATX	150	86	140 - 210
TFX	85	65	175
SFX	100	63.5	125
SFX-L	130	63.5	125

Common PSU sizes

WxH Standard
More power can mean more depth...
600W 140mm depth, 1600W 21mm depth

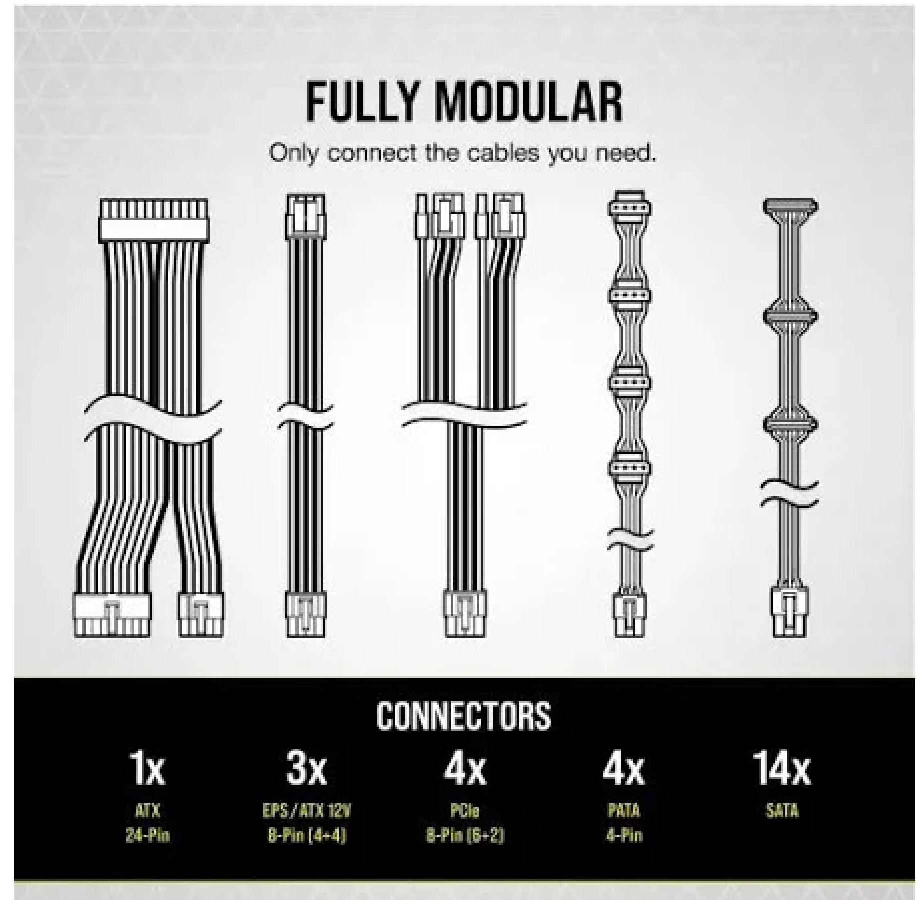
TFX

SFX-L

SFX

Modularity

Modular PSU



Most recent ATX are modular

Choose your cables

Tidier build

Non-Modular PSU

Cables pre-attached, not changeable

Can be cheaper

Less tidy...

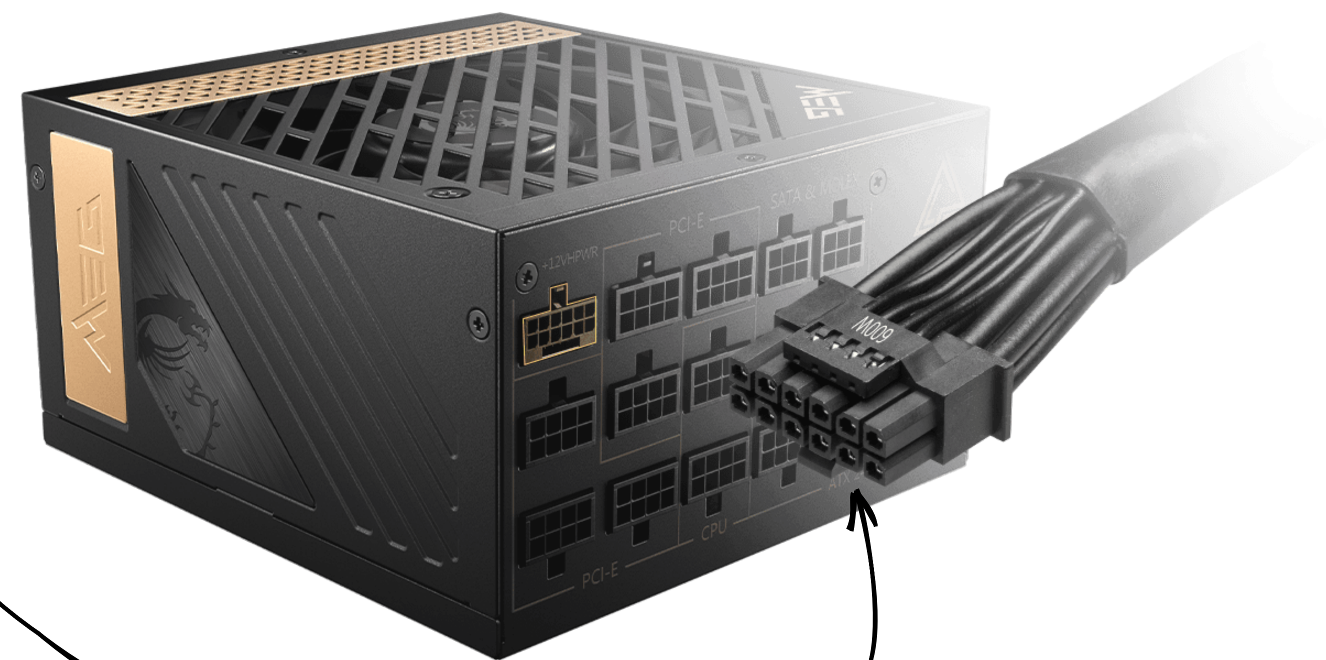
ATX 3.0

New standard released 2022, first change in 20 years

Comes with a 12+4 (16) pin 12VHPWER PCIe 5.0 connector for the very latest and next generation GPUs

Not necessary yet (2023) unless your GPU needs one

Future proof option



PSU Recommendations

Go with a known, established brand

Seasonic, Corsair, Asus, Thermaltake, MSI, EVGA, Coolermaster or Gigabyte

Don't go on customer reviews for a budget make, reviews are usually not long term!

Look at warranty for longevity

My preferred level, good efficiency, good quality, reasonable price point.

Platinum and Titanium for always on and expensive electricity costs, gains are marginal

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Efficiency Certification @ 100% Load